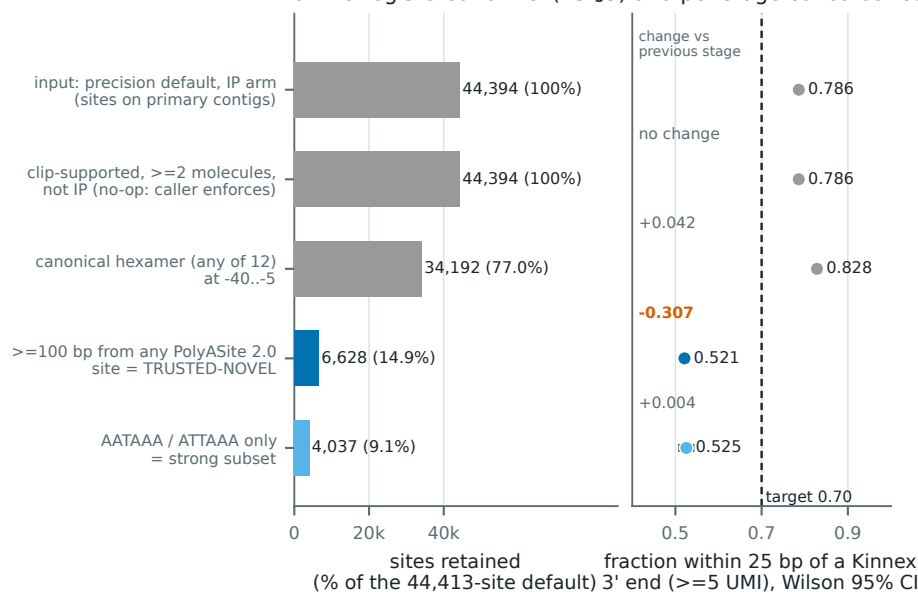


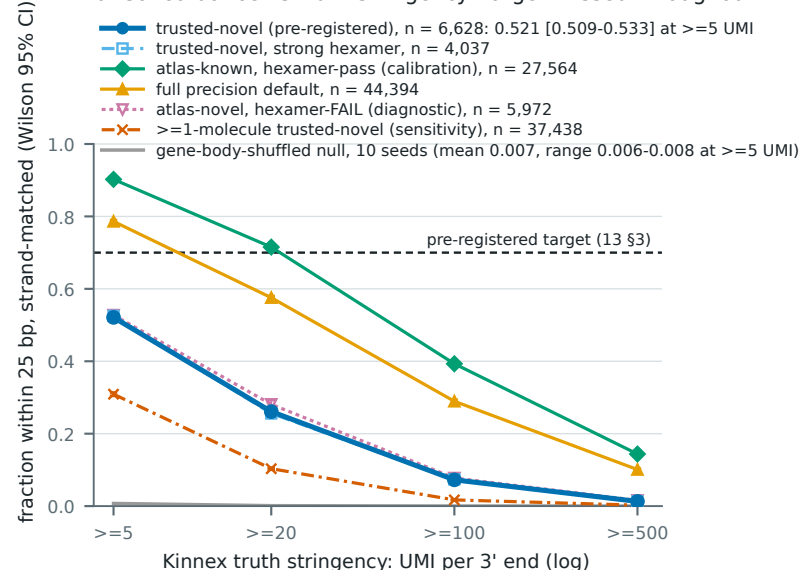
Pre-registered trusted-novel definition: 52% long-read concordance, target 70% not met

PBMC 10k v3, PeakATail precision default (IP arm, code 4efeb125; 44,394 sites on primary contigs) -> 6,628 trusted-novel (14.9%, 3,350 genes; 4,037 with AATAAA/ATTAAG). Trusted-novel = clip-supported, >=2 molecules, not internal priming, canonical hexamer at -40..-5, >=100 bp from any PolyASite 2.0 site (13 §3, committed before any number existed). Truth = Kinnex x3p poly(A)-verified 3' ends; hit = within 25 bp on the same strand. Result: 0.521 [0.509-0.533] at >=5 UMI (3,454 / 6,628); 0.261 / 0.073 / 0.013 at >=20 / >=100 / >=500 UMI.

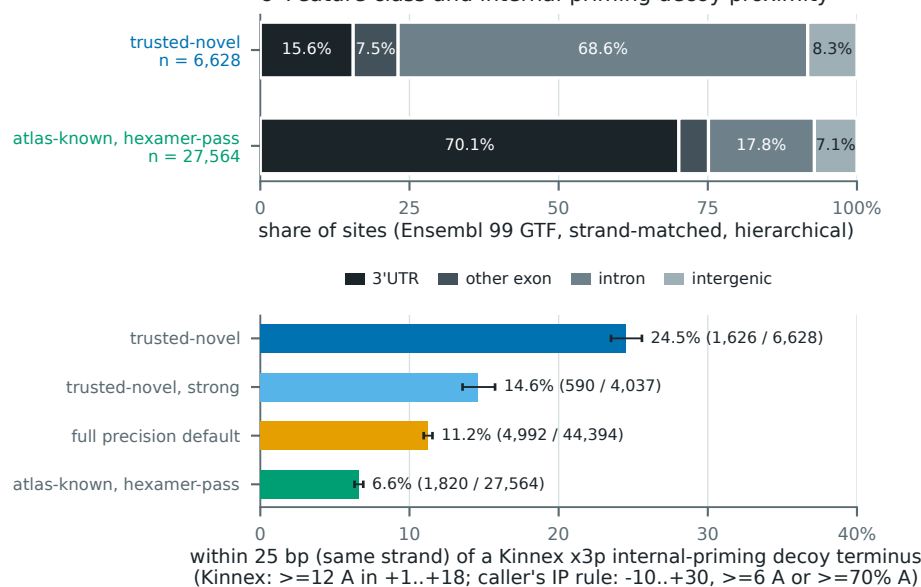
a Pre-registered funnel (13 §3) and per-stage concordance



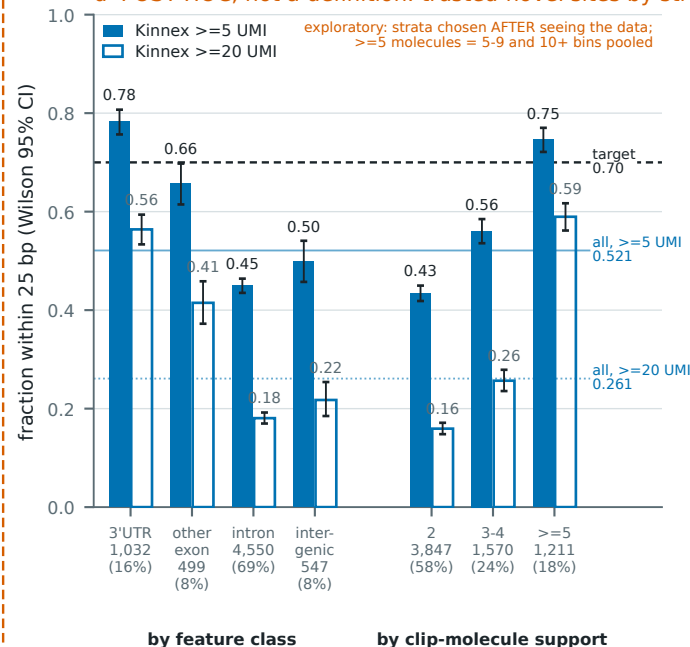
b Concordance vs truth stringency: target missed throughout



c Feature class and internal-priming decoy proximity



d POST HOC, not a definition: trusted-novel sites by stratum



Truth caveats: both Kinnex sets come from donors other than the PBMC 10k v3 donor, so a site used in this sample but absent (or <5 UMI) in the long-read donor counts as a miss -- the atlas-known hexamer-pass complement scores 0.902 under the same truth, so donor mismatch cannot explain the gap. Robustness (verifier, not part of the pre-registered metric): GEM-X truth 0.554 (3,670/6,628); x3p+GEM-X pooled 0.623 at 25 bp, 0.698 at 50 bp, 0.739 at 100 bp -- the pre-registered 25-bp metric is missed under every truth choice. Null = sites shuffled within gene bodies (10 seeds; mean 0.0074 at >=5 UMI, enrichment 71x, empirical p at the 0.091 = 1/11 floor): the sites are far from random, but the target is an absolute fraction. Funnel stages clip / >=2 molecules / not-IP remove nothing because the caller already enforces them (re-deriving IP from the genome would flag <=3%); PolyA_DB could not be used (hg19 only). The hexamer stage adds +0.042 on the whole set but nothing among atlas-novel sites (hexamer-FAIL 0.528 vs trusted-novel 0.521); the atlas-novelty stage costs -0.307. Panel d is exploratory: every stratum was chosen after seeing the data, no stratum is a definition, and none has been validated on held-out data (candidate v2: 3'UTR-restricted and/or >=5 molecules and/or a Kinnex-style IP rule, Stage-1d). Single PBMC donor, single CellRanger BAM. Sources: results/reliability/trusted_novel_final_pbmc/(REPORT_PROVISIONAL.md §1-9, verifier-reproduced), manuscript/16_trusted_novel_kinnex.md (verified FIXED). Every plotted value: results/figures/manuscript/trusted_novel_funnel*.tsv.